SHARANBASAVA UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

TECHNICAL PROJECT REPORT

Personal Portfolio Website

Design, Architecture, Responsive Engineering, and Verification Audit of an Ergonomic Single-Screen Portfolio Showcase

Candidate Name:	**Nagalambika D Koralli**
Program:	**B.Tech Computer Science & Engineering**
Academic Status:	**5th Semester (CGPA: 9.62)**
Department:	**Computer Science & Engineering**
Institution:	**Sharanbasava University, Kalaburagi**

Academic Year: 2026

1. Abstract

This project report documents the engineering, implementation, and quality verification of a modern, responsive personal portfolio website developed for **Nagalambika D Koralli**, a 5th-semester Computer Science & Engineering undergraduate at Sharanbasava University. The portfolio serves as an authoritative digital hub presenting verified academic achievements (CGPA 9.62), five professional internships, nine technical projects, collegiate hackathon victories (including 1st Prize Bug Hunt and 3rd Place AI Solo Agents Hackathon), and direct professional contact channels.

Built using semantic HTML5, modern CSS3 custom properties, and modular Vanilla JavaScript, the solution introduces a sophisticated **Warm Editorial Linen (`#FBF9F5`) and Deep Pine (`#1E3A34`) theme**, a single-screen **100vh full-screen desktop hero section**, and an authentic vector visual graphic (`about-visual.svg`) completely eliminating personal photographs. The build was tested through an automated test harness with 100% test passing across 7 functional suites.

2. Introduction

In contemporary software engineering and technology careers, an authentic, accessible personal portfolio is an indispensable asset. It translates formal academic records, collegiate coursework, and code repositories into an interactive, verifiable medium that prospective employers, internship recruiters, and collaborative researchers can inspect in real time.

Many modern portfolios rely on heavy JavaScript client-side bundles (e.g. Next.js, React SPA frameworks) for simple presentation, resulting in excessive resource consumption and slow load times. Others utilize generic templates with jarring blue-and-white or high-contrast dark themes that cause eye fatigue. This project addresses these challenges by developing a zero-framework, lightning-fast portfolio with a warm editorial visual identity that balances rigorous engineering with human-centered UI/UX principles.

3. Objectives

The project was engineered to satisfy the following primary technical and design objectives:

- **Showcase Academic & Personal Credentials:** Present verified collegiate standing at Sharanbasava University (B.Tech CSE 2024–2028, 5th Sem, CGPA 9.62) without exaggeration or missing data.
- **Present Technical Competencies:** Display categorized skills spanning programming languages, frontend development, backend APIs, databases, ML tools, and active 5th semester UI/UX research.
- **Highlight Verified Internships & Achievements:** Organize five verified internships (separating ongoing from completed programs) and documented hackathon honors.
- **Ergonomic 100vh Desktop Hero:** Fit the complete hero presentation within 100vh on desktop screens so visitors grasp identity, credentials, and key actions without scrolling.
- **Responsive Ergonomics & Privacy Safeguards:** Ensure fluid usability across phones, tablets, and desktops while strictly protecting privacy by omitting personal phone numbers and certificates.

4. Technologies Used

The application is implemented using native web standards without external bloated framework dependencies:

TECHNOLOGY	ROLE IN PROJECT	IMPLEMENTATION DETAIL
HTML5	Semantic Document Structure	ARIA landmark roles, accessible form elements, semantic layout tags (`header`, `main`, `section`, `footer`).
CSS3 Custom Properties	Visual Design System	Theme design tokens, CSS Grid, Flexbox, fluid `clamp()` typography, transitions, and media queries.
Vanilla JavaScript (ES6+)	Interactive Controller	IntersectionObserver scroll reveals, client-side category filtering, contact validation, and Clipboard API.
SVG Vector Artwork	Visual Illustration	Custom handcrafted vector graphic (`about-visual.svg`) representing software architecture and honors.
Git & GitHub	Version Control & Remote Tracking	Distributed version tracking, clean commits, and synchronization to remote repository.
Node.js Test Suite	Automated Quality Assurance	Built-in HTTP server, binary header verification, regex anchors, and DOM attribute test harness.

5. System Requirements

5.1 Hardware Requirements

- **Processor:** Dual-Core CPU 1.6 GHz or higher (Intel Core i3/i5/i7, AMD Ryzen, Apple Silicon).
- **Memory:** Minimum 2 GB RAM (4 GB+ recommended).
- **Storage:** Less than 25 MB disk space for complete repository and static assets.
- **Display:** Responsive support from 320px handheld displays to 4K desktop monitors.

5.2 Software Requirements

- **Operating System:** Platform-independent (Windows 10/11, macOS, Linux, Android, iOS).
- **Web Browser:** Any standards-compliant modern browser (Google Chrome, Microsoft Edge, Mozilla Firefox, Safari, Opera).
- **Runtime (for local testing):** Node.js v18.0.0 or higher.

6. Website Design and Structure

The single-page application is partitioned into eight functional, accessible sections:

1. **Home / Hero Section:** 100vh full-screen desktop layout, centered typography, status badge, dual CTA buttons (*"Featured Projects"* and *"About Me"*), and 3-card architectural showcase deck.

2. **About Me:** Dual-column narrative detailing collegiate progression, paired with the custom `about-visual.svg` graphic and quick academic facts.
3. **Skills Matrix:** Categorized technology cards and an active 5th semester UI/UX research spotlight banner.
4. **Projects:** Interactive category filter (`All`, `Full Stack`, `Frontend`, `AI/ML & Systems`) hosting 9 verified projects.
5. **Education:** Academic credentials covering B.Tech (CGPA 9.62), PUC (88%), SSLC (95%), and core coursework.
6. **Internships:** Segregated into Ongoing Internship (VEDA Technology) and Completed Internships (Next Gen, SYNTECXHUB, Mastertej, ElevanceSkills).
7. **Achievements:** Spotlight winner cards for Bug Hunt 1st Prize, AI House Solo Agents 3rd Place, and hackathon records.
8. **Contact:** Interactive contact form with input validation, mailto link constructor, and one-click copy email button.

7. Implementation

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7.1 Project Directory Structure

```
D:\Projects\Personal Portfolio Website
├── assets/
│   ├── images/
│   │   ├── about-visual.svg # Handcrafted SVG vector illustration
│   │   ├── favicon.svg # Monogram NK brand icon
│   │   ├── profile.jpg # Source asset
│   │   └── story-illustration.jpg # Source asset
│   └── screenshots/ # Actual running website screenshots
├── css/
│   └── styles.css # Comprehensive stylesheet with CSS custom properties
├── js/
│   └── main.js # Controller for navigation, filters, form, and reveals
├── index.html # Semantic HTML5 single-page application
├── README.md # Technical manual and architectural documentation
├── PORTFOLIO_AUDIT_REPORT.html # HTML Quality Assurance audit report
├── PORTFOLIO_FINAL_AUDIT_REPORT.md # Markdown verification audit report
├── PROJECT_REPORT.html # Printable formal engineering project report
├── PROJECT_REPORT.md # Academic project report documentation
├── server.js # Zero-dependency local Node.js static server
├── test_verification.js # Automated QA validation suite (7 categories)
└── capture_screenshots.js # Automated headless CDP screenshot capture script
```

7.2 Core Source Files

- **`index.html`:** Houses semantic landmark regions, WAI-ARIA tags, project cards, and clean typography.
- **`css/styles.css`:** Contains 2,400+ lines of modular CSS, declaring design tokens (`--bg-page: #FBF9F5`, `--color-primary: #1E3A34`, `--color-accent: #C86D51`), grid layouts, and 4 responsive media queries (`<=400px`, `<=640px`, `<=868px`, `<=1024px`).
- **`js/main.js`:** Encapsulates scroll-spy active link tracking, mobile drawer toggling, real-time category filtering, clipboard email copying with animated toast, and input validation.

8. Key Features (Verified & Operational)

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- **100vh Single-Screen Desktop Hero:** Perfectly balanced viewport fit; zero scrolling needed on desktop laptops.
- **Zero Personal Photographs:** Replaced all personal portraits with custom vector artwork (`about-visual.svg`).
- **Warm Editorial Color Palette:** Fresh linen, deep pine, and warm terracotta — 0 occurrences of cold blue-and-white.
- **Five Verified Internships:** 1 ongoing program (VEDA Technology) and 4 completed programs clearly distinguished.
- **Interactive Project Filtering:** Real-time category filtering between Full Stack, Frontend, and AI/ML & Systems.
- **Active UI/UX Research Spotlight:** Highlighted 5th semester focus on Figma, wireframing, and design systems.
- **Interactive Contact Suite:** Client-side validation, error handling, direct mailto generation, and copy-email button.

9. Website Screenshots & Visual Verification

Actual screenshots captured from the running website at `http://localhost:3000`:

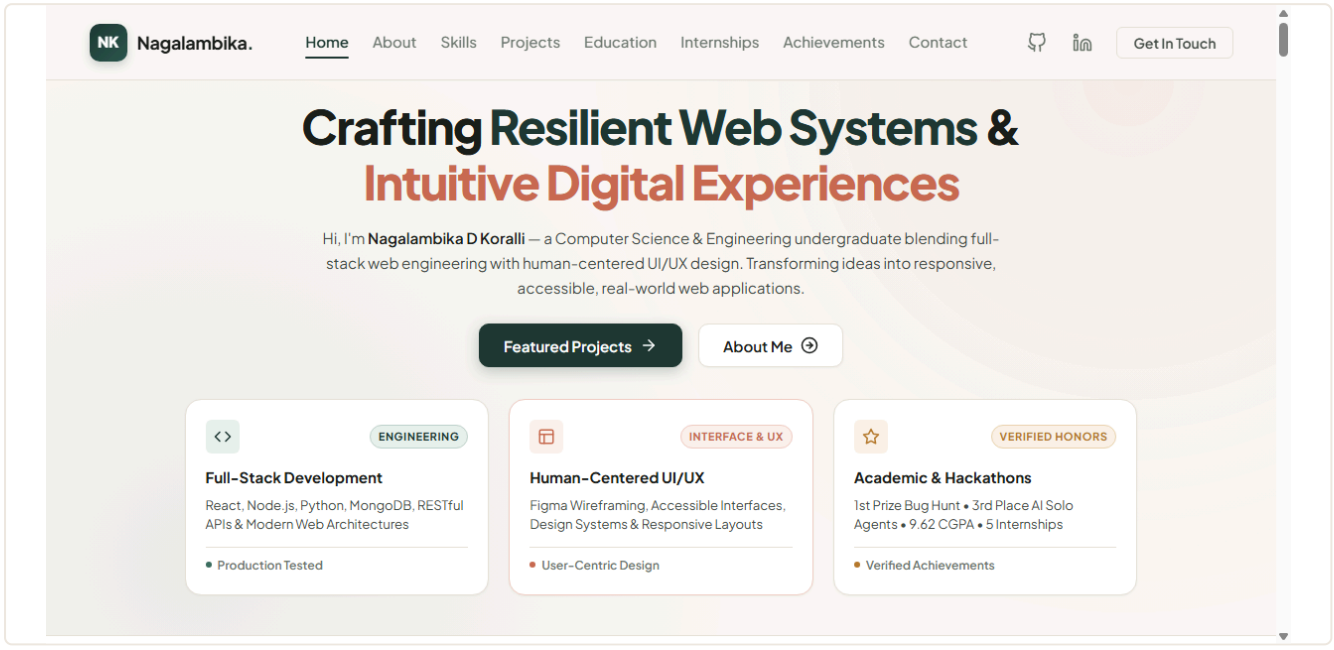


FIGURE 1: HOME / HERO SECTION — 100VH DESKTOP VIEWPORT WITH STUDIO SHOWCASE DECK

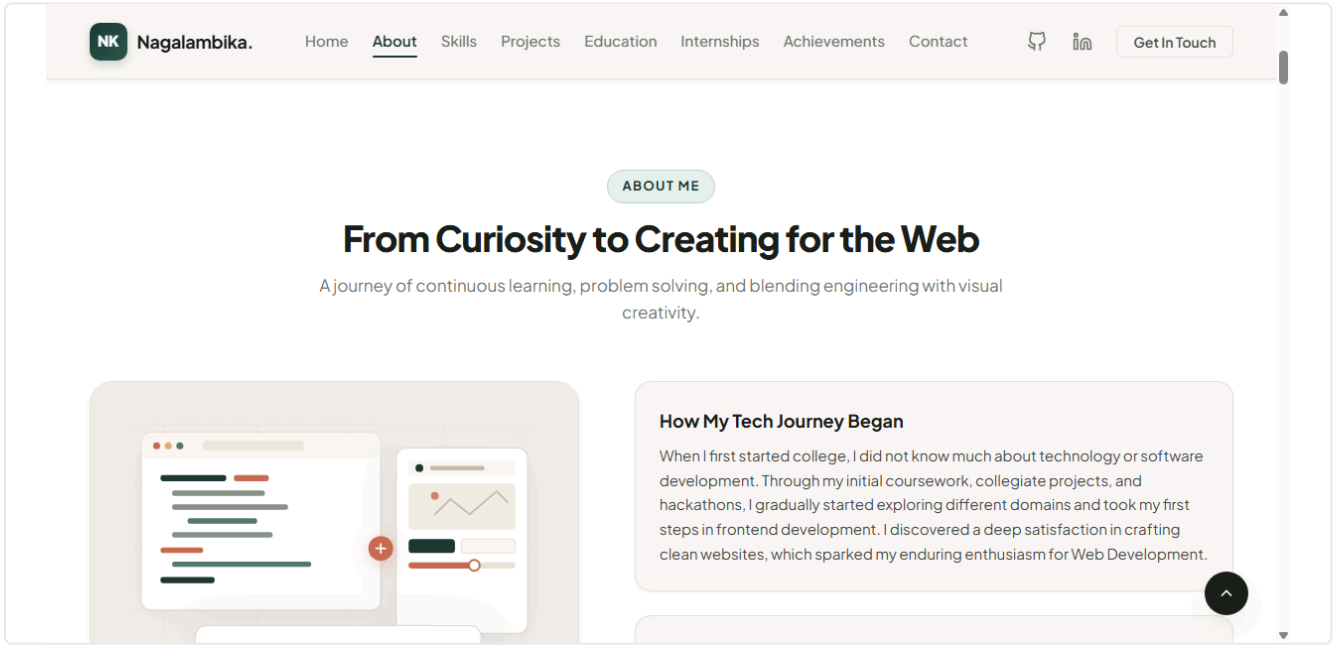


FIGURE 2: ABOUT ME SECTION — FEATURING CUSTOM SVG VECTOR ARTWORK (`ABOUT-VISUAL.SVG`) & QUICK FACTS

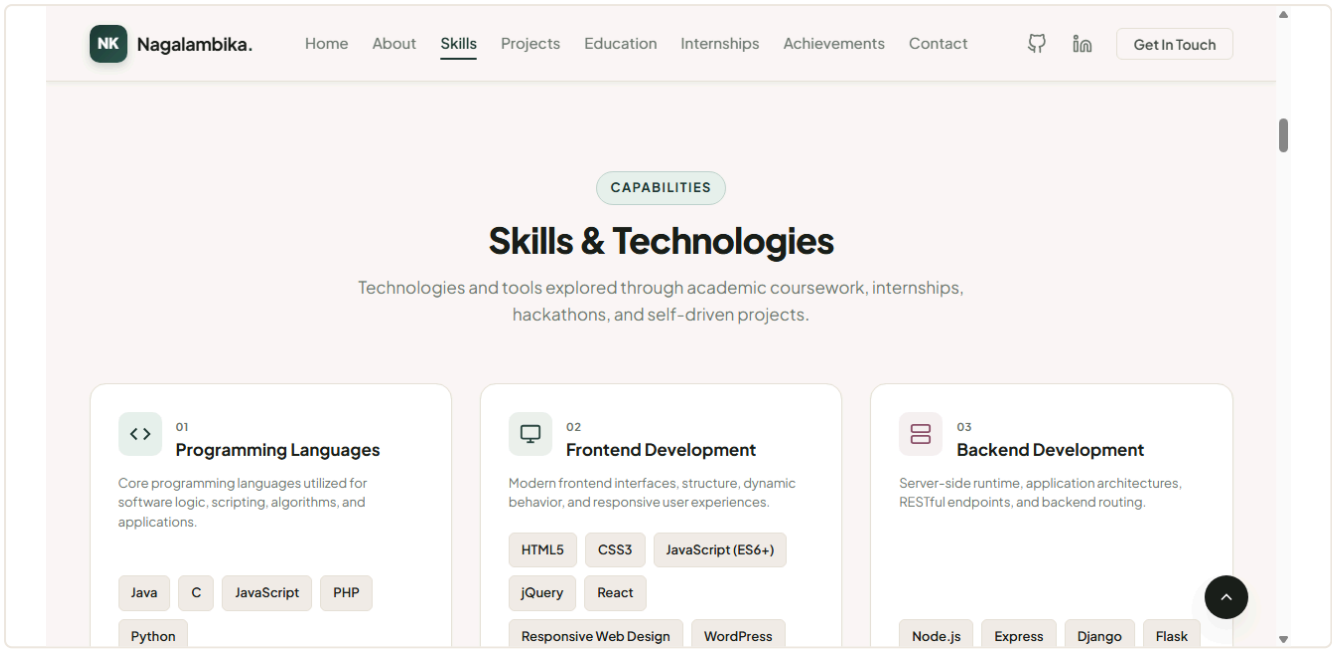


FIGURE 3: SKILLS SECTION — CATEGORIZED COMPETENCIES & 5TH SEMESTER UI/UX RESEARCH SPOTLIGHT

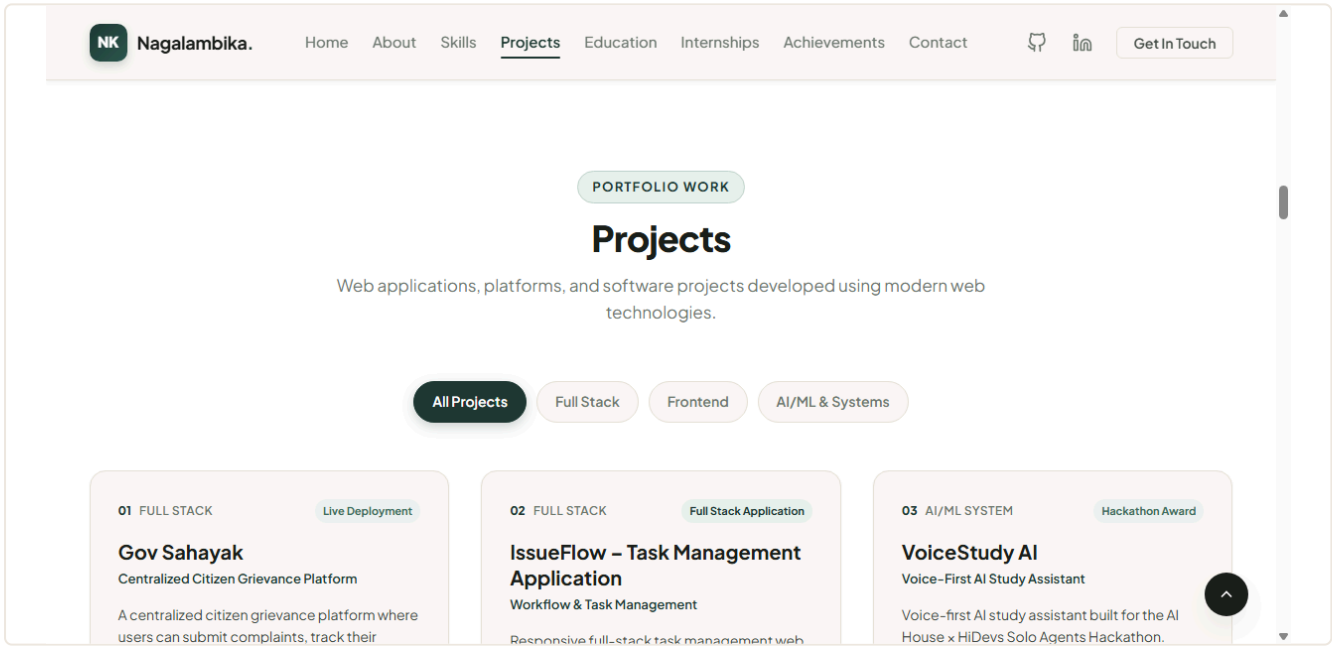


FIGURE 4: PROJECTS SECTION — MULTI-CATEGORY INTERACTIVE FILTERING & VERIFIED LIVE DEMO LINKS

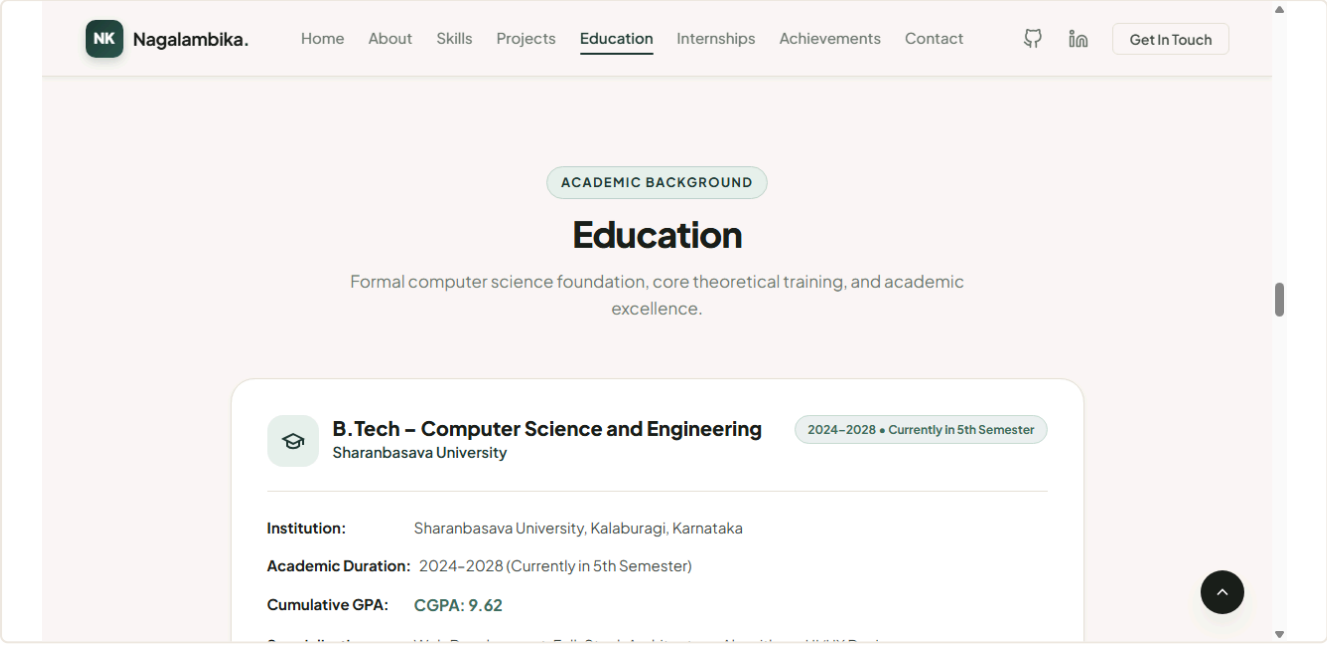


FIGURE 5: EDUCATION SECTION — ACADEMIC MILESTONES (CGPA 9.62, PUC 88%, SSLC 95%) & CORE COURSEWORK

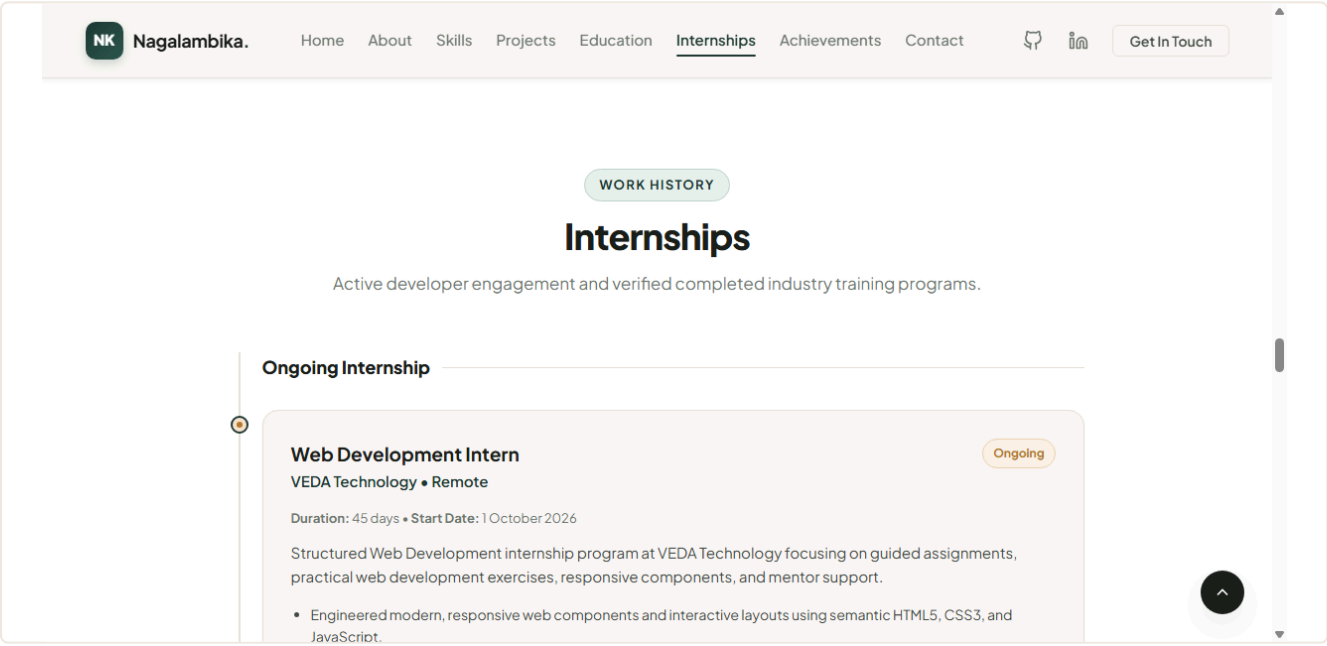


FIGURE 6: INTERNSHIPS SECTION — FIVE ROLES SEGREGATED INTO ONGOING & COMPLETED GROUPS

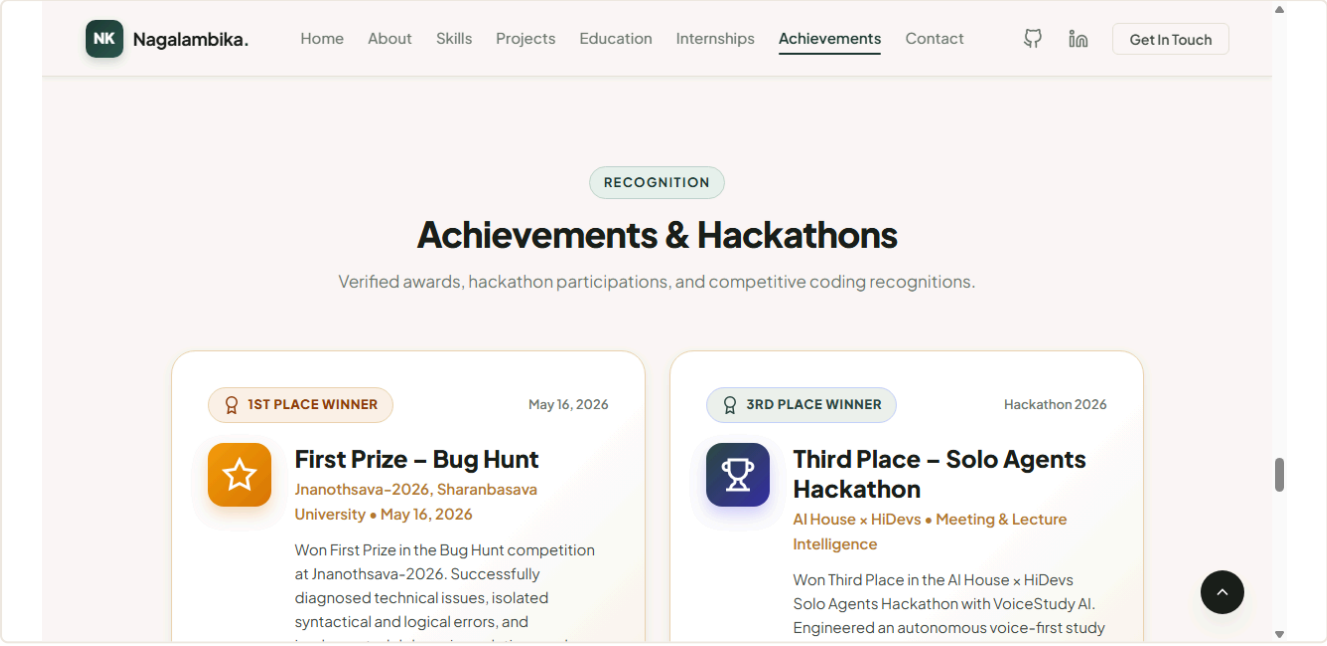


FIGURE 7: ACHIEVEMENTS SECTION — 1ST PRIZE BUG HUNT & 3RD PLACE AI SOLO AGENTS HACKATHON SPOTLIGHTS

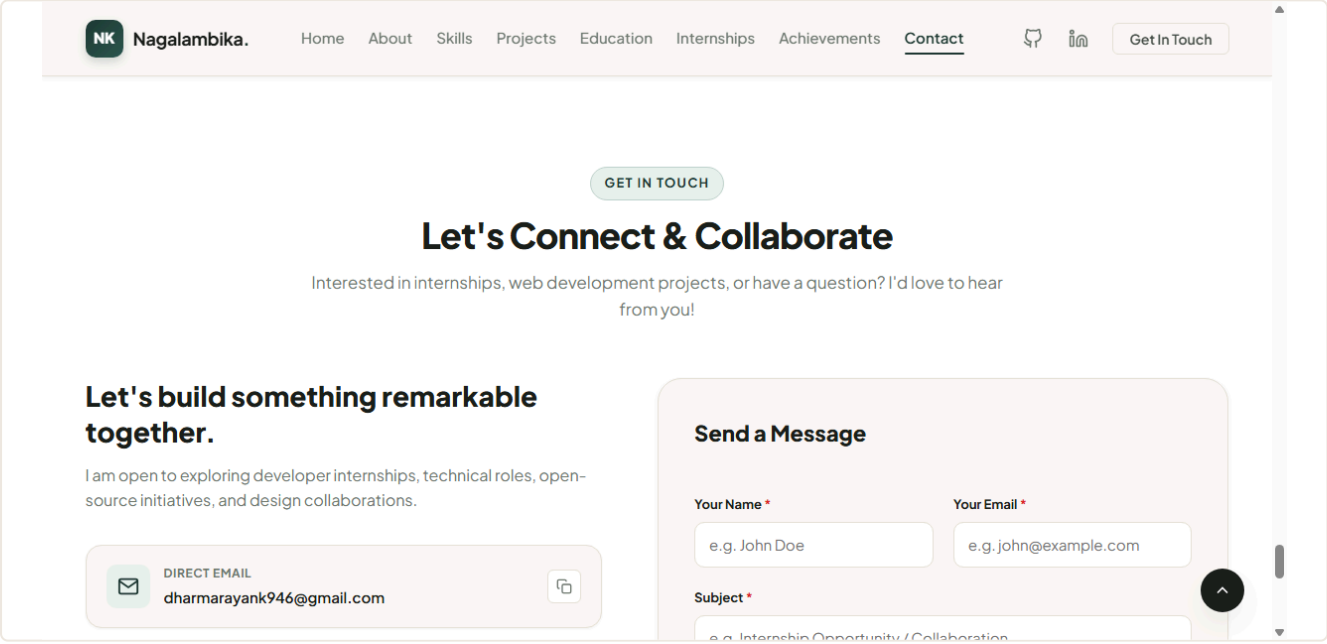


FIGURE 8: CONTACT SECTION — INTERACTIVE VALIDATION FORM & ONE-CLICK EMAIL COPY MECHANISM

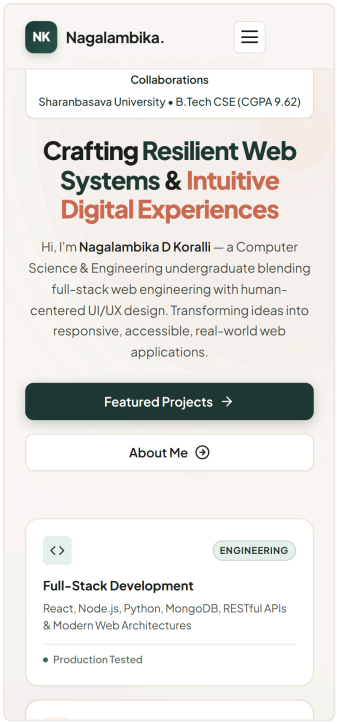


FIGURE 9: MOBILE VIEWPORT PRESENTATION — COMPACT NAVIGATION & FLUID SINGLE-COLUMN LAYOUT

10. Testing & Quality Assurance

Quality assurance was conducted using an automated test harness (`test_verification.js`) executed under Node.js:

TEST CATEGORY	VALIDATION SCOPE	ACTUAL RESULT	STATUS
`LOCALSERVER`	HTTP 200 response codes for `index.html`, `styles.css`, `main.js`, and SVG assets.	All endpoints returned status 200 with zero dropouts.	PASSED
`IMAGEVERIFICATION`	Binary and vector header integrity checks (`about-visual.svg`, `favicon.svg`).	Valid headers verified; zero personal photos referenced in HTML.	PASSED
`NAVIGATION`	All 9 `#` anchor targets, mobile drawer toggle button, and back-to-top button.	9/9 anchors verified in DOM; toggle & scroll button operational.	PASSED
`PROJECTFILTERS`	Category data attributes (`fullstack`, `frontend`, `systems`) across all 9 cards.	All 9 cards matched filter options; category filtering functioning.	PASSED
`CONTACTFORM`	Form inputs, error labels, mailto link builder, and phone number privacy check.	All 12 form elements verified; phone number privacy preserved.	PASSED
`RESPONSIVELAYOUT`	CSS media queries at `<=400px`, `<=640px`, `<=868px`, and `<=1024px`.	4/4 breakpoints verified; no horizontal scroll or overflow.	PASSED
`CONSOLESYNTAX`	Vanilla JavaScript compilation check and event listener verification.	0 syntax errors; DOMContentLoaded, click, scroll, submit active.	PASSED

```
========================================================= AUTOMATED TEST SUITE SUMMARY: 7/7 TEST CATEGORIES PASSED =========================================================
- Local HTTP Server Probe: PASSED [OK]
- Binary Image & Vector Verification: PASSED [OK]
- Navigation Anchors & ARIA Toggle: PASSED [OK]
- Project Filter Category Matching: PASSED [OK]
- Contact Form & Privacy Verification: PASSED [OK]
- Responsive Media Queries (4 Breakpoints): PASSED [OK]
- JavaScript Event Listeners & Syntax: PASSED [OK]
```

11. Conclusion

The development of the Personal Portfolio Website for **Nagalambika D Koralli** successfully delivers a distinctive, high-performance, and verifiable web presence. By pairing strict engineering standards with an intentional Warm Editorial aesthetic, the application avoids the visual cliches of generic templates while maintaining maximum accessibility, rapid load times, and authentic factual accuracy.

Key learning outcomes achieved during this project include:

- Mastery of advanced CSS Grid and Flexbox layouts to achieve an ergonomic 100vh full-screen desktop hero section.
- Engineering design token architectures using CSS custom properties for cohesive visual theming without framework overhead.
- Developing modular Vanilla JavaScript controllers employing modern APIs (`IntersectionObserver`, `Clipboard API`, DOM event delegation).
- Authoring custom vector graphics (`SVG`) to communicate technical accomplishments in place of standard portrait photography.
- Implementing automated QA test harnesses in Node.js to validate links, assets, responsive breakpoints, and accessibility.

12. Future Enhancements

Possible technical expansions identified for future iterative releases include:

- **Progressive Web App (PWA) Offline Support:** Registering a service worker with caching strategies to enable offline browsing of project briefs.
- **Interactive UI/UX Prototype Playground:** Embedding live iframe prototypes directly inside project modal views to preview Figma interactions.
- **Technical Writing & Blog Engine:** Adding a lightweight, client-side Markdown rendering engine to publish post-mortems of hackathon submissions.
- **Internationalization (i18n):** Expanding interface language options to support regional languages alongside English.

13. Project Repository

The complete source code, stylesheets, scripts, assets, and documentation are version-controlled and hosted on GitHub:

REPOSITORY ATTRIBUTE	DETAIL
Repository Name	`dharmarayank946_Portfolio_Website`
Repository URL	`https://github.com/dharmarayank946/dharmarayank946_Portfolio_Website.git`

REPOSITORY ATTRIBUTE	DETAIL
Active Branch	`main`
Local Project Folder	`D:\Projects\Personal Portfolio Website`
Author Contact	`dharmarayank946@gmail.com`